

Myanmar Earthquake & Dam Risk Report

1. Executive Summary

This report covers 9,420 earthquake events in Myanmar from 1970-07-29 to 2026-07-04. The largest event was M7.7 on 1988-11-06. The Gutenberg-Richter b-value is 1.050 with magnitude of completeness $M_c=4.7$.

Total Events: 9,420

Date Range: 1970-07-29 to 2026-07-04

Max Magnitude: M7.7

Mean Depth: 24.8 km

b-value: 1.050

a-value: 7.720

M_c (completeness): 4.7

2. Earthquake Map

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3. Seismic Clustering Analysis

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4. Dam Seismic Risk Assessment

Risk assessment for 254 dams using PGA estimation from the M7.7 mainshock, fault proximity, and structural exposure scoring.

Critical Risk: 45 dams

High Risk: 134 dams

Moderate Risk: 61 dams

Low Risk: 14 dams

Top 30 Highest Risk Dams:

Name	River	PGA(g)	Dist	Fault	Risk	Grade			
Dam near 21.74N 95.85E		nan	0.3656	7.4 km	10.0	Critical			
Dam near 22.09N 95.86E		nan	0.4019	5.7 km	10.0	Critical			
Dam near 21.77N 95.89E		nan	0.4626	2.7 km	10.0	Critical			
Dam near 21.47N 95.87E		nan	0.3944	6.0 km	9.7	Critical			
Dam near 20.85N 96.03E		nan	0.3478	8.3 km	8.8	Critical			
Meikhtila		nan	0.3525	8.2 km	8.8	Critical			
Dam near 21.34N 95.81E		nan	0.2956	12.2 km	8.8	Critical			
Natkar		nan	0.3593	7.9 km	8.7	Critical			
Dam near 21.37N 95.80E		nan	0.2830	13.4 km	8.7	Critical			
Dam near 20.72N 95.90E		nan	0.4000	5.6 km	8.7	Critical			
Dam near 21.04N 95.83E		nan	0.3122	10.9 km	8.7	Critical			
Dam near 21.05N 95.83E		nan	0.3105	11.0 km	8.7	Critical			
Dam near 20.47N 96.08E		nan	0.3252	9.9 km	8.6	Critical			
Dam near 20.60N 95.90E		nan	0.3681	7.4 km	8.6	Critical			
Taungnyo	Taungnyo	0.5047	1.2 km	8.6	Critical				
Nat Thar Taw	Nat Thar Taw	0.2265	20.2 km	8.5	Critical				
Myotha	Ku Hta	0.2211	20.6 km	8.5	Critical				
Ta Nai Hka	Ta Nai Hka	0.4441	3.9 km	8.2	Critical				
Zeedaw	Nyaung Oat	0.2633	15.3 km	8.2	Critical				
Dam near 17.55N 96.37E		nan	0.3298	9.6 km	8.2	Critical			
Thaphan Chaung	Thaphan Chaung	0.3041	11.5 km	8.2	Critical				
Male-Nattaung	Male-Nattaung	0.2327	18.6 km	8.1	Critical				
Nyaunggone		nan	0.2795	13.7 km	7.9	Critical			
Paunggadaw	Paunggadaw	0.1673	33.9 km	7.9	Critical				
Dam near 20.73N 95.83E		nan	0.2809	13.6 km	7.9	Critical			
Kun Chaung	Kun Chaung	0.5304	0.9 km	7.7	Critical				
Kintat	Mu	0.2235	21.0 km	7.7	Critical				
North Pinle	Pinle	0.1599	36.1 km	7.7	Critical				
Phyu Chaung	Pyu Chaung	0.5211	1.3 km	7.6	Critical				
Kabaung	Kabaung Chaung	0.3302	9.2 km	7.6	Critical				

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5. Aftershock Forecast

Modified Omori law parameters fitted to aftershocks of M7.7 mainshock.

K (productivity): 34.92

c (time offset): 16.408 hours

p (decay exponent): 0.772

Expected M $\geq$ 3 (7d): 4

Expected M $\geq$ 3 (30d): 18

Expected M $\geq$ 3 (90d): 53

Note: These are statistical estimates based on the Omori decay law. Actual aftershock rates may vary due to local geological conditions.

6. Data Sources

- Earthquake data: mmeq.akze.net API (USGS/ISC sourced)
- Dam data: Open Development Mekong (IFC/WLE), CC BY-SA 4.0
- Fault lines: USGS/Plate boundary data
- PGA estimation: Abrahamson & Silva (2008) NGA-West1 GMPE
- Report generated by mmeq-opendata toolkit